DATA CLEANING USING



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1. Introduction

Data cleaning is an essential step in the data analytics process. Raw datasets often contain duplicate records, missing values, inconsistent naming conventions, unwanted whitespace, and incorrectly formatted data types. These issues can affect the accuracy of analysis and lead to misleading business insights.

This project focuses on cleaning a layoffs dataset using MySQL. The objective is to identify and address data quality issues, standardize inconsistent values, investigate missing information, and create a clean dataset suitable for further analysis.

The project follows a structured workflow, preserving the original imported dataset while performing cleaning operations on a separate working table.

2. Project Objectives

The primary objectives of this project are:

1. Import the raw layoffs dataset into MySQL.
2. Preserve the original data before performing cleaning operations.
3. Identify and remove duplicate records.
4. Standardize company names by removing unwanted whitespace.
5. Identify and address inconsistencies in the industry column.
6. Standardize country names.
7. Convert the date column into the appropriate data type.
8. Investigate missing industry information and recover values where possible.
9. Examine missing values in the layoff-related columns.
10. Create a clean dataset containing records with complete layoff information.

3. Tools and Technologies

Tool or technology	Purpose
MySQL Workbench	Database management and data cleaning
SQL	Data inspection, transformation, filtering, and validation
CSV	Source data format

4. Dataset Overview

The project began with a CSV file containing company layoff information. The dataset was imported into MySQL for cleaning.

The original dataset contained 2,361 records and nine columns.

The columns used during the cleaning process were:

Column	Description
company	Name of the company reporting layoffs
location	Location associated with the company or layoff event

industry	Industry in which the company operates
total_laid_off	Number of employees laid off
percentage_laid_off	Percentage of the workforce laid off
date	Date associated with the layoff event
stage	Company funding or development stage
country	Country associated with the company or layoff event
funds_raised_millions	Funding raised, expressed in millions

5. Data Cleaning Methodology

The cleaning process was divided into several steps. The original imported table was preserved, and transformations were performed on a separate working table.

Step 1: Import the Dataset into MySQL

The CSV file was imported into MySQL as the layoffs table.

The initial row count was verified.

```
SELECT COUNT(*)
FROM layoffs;
```

Finding: The imported dataset contained 2,361 rows.

Step 2: Create a Working Copy

To protect the original dataset, a separate table named raw_layoffs was created.

```
CREATE TABLE raw_layoffs AS
SELECT *
FROM layoffs;
```

This provided a working copy for data cleaning while retaining the original imported table as a reference.

Step 3: Identify and Remove Duplicate Records

Duplicate records were identified by comparing all nine columns.

```
SELECT
    company,location,industry, total_laid_off, percentage_laid_off,
    `date`,stage,country,funds_raised_millions,
    COUNT(*) AS duplicate_count
FROM raw_layoffs
GROUP BY
    company,location,industry,total_laid_off,percentage_laid_off,`date`,
    stage,country,funds_raised_millions
HAVING COUNT(*) > 1;
```

The investigation identified five groups of exact duplicate records:

Company	Duplicate records
Yahoo	2
Wildlife Studios	2
Cazoo	2
Casper	2
Hibob	2

Each group contained two identical records. Therefore, five excess records were identified.

The Yahoo records were additionally verified using:

```
SELECT *
FROM layoff_db.raw_layoffs
WHERE company = 'Yahoo';
```

Both records contained identical values across all nine columns.

The duplicate records were subsequently removed using a unique row identifier and ROW_NUMBER() to retain one copy of each distinct record.

Result:

Metric	Count
Original records	2,361
Duplicate records removed	5
Records remaining	2,356

The working dataset was reduced to 2,356 records.

Step 4: Remove Unwanted Whitespace from Company Names

A query was used to identify company names containing leading spaces.

```
SELECT company
FROM raw_layoffs
WHERE company LIKE ' %';
```

The following company names were identified:

- E Inc.
- Included Health

To standardize company names, the following query was executed:

UPDATE raw_layoffs

SET company = TRIM(company);

The TRIM() function removes leading and trailing spaces from company names.

Result: Unwanted whitespace was removed from the company column, improving consistency in company names and supporting more reliable grouping and filtering.

Step 5: Identify and Standardize Industry Names

During the inspection of the industry column, three data quality issues were identified.

Issue 1: Missing values

One NULL value and one blank value were initially identified.

Issue 2: Inconsistent industry naming

The following values represented the same industry:

- Crypto
- Crypto Currency
- CryptoCurrency

To standardize these values, the following query was executed:

```
UPDATE raw_layoffs
SET industry = 'Crypto'
WHERE industry LIKE 'Crypto%';
```

This changed the three identified variations to the consistent industry name Crypto.

Original value	Standardized value
Crypto	Crypto
Crypto Currency	Crypto
CryptoCurrency	Crypto

Result: Inconsistent cryptocurrency industry labels were standardized, allowing these records to be grouped under one industry category.

Step 6: Standardize Country Names

During the inspection of the country column, four records contained United States. instead of United States.

The trailing period caused these records to appear as a distinct country value.

Initially, a direct replacement was considered. However, the following more general query was used:

```
UPDATE raw_layoffs
SET country = TRIM(TRAILING '.' FROM country);
```

This removes trailing periods from country values.

Result: The four identified United States. values were standardized to United States. The query also addressed any other country values ending with a period.

Step 7: Standardize the Date Column

The date column was initially stored as text, making it unsuitable for reliable date-based operations without conversion.

The following query was used to convert the existing text values into date values:

```
UPDATE raw_layoffs
```

```
SET `date` = STR_TO_DATE(`date`, '%m/%d/%Y');
```

The format string %m/%d/%Y represents the original month/day/year format.

For example:

Original representation	Standardized representation
2/9/2023	2023-02-09
11/28/2022	2022-11-28
6/7/2022	2022-06-07

After converting the values, the column's data type was changed to DATE.

ALTER TABLE raw_layoffs

MODIFY COLUMN `date` DATE;

Result: The date column was converted to the appropriate data type, enabling chronological sorting, date filtering, and time-based analysis.

Step 8: Investigate Missing Industry Values

Further investigation identified four companies with missing industry information.

The following query was used:

```
SELECT
    company,
    industry
FROM raw_layoffs
WHERE industry IS NULL
   OR industry = '';
```

The four companies identified were:

Company	Initial finding
Airbnb	Missing industry
Bally's Interactive	Missing industry
Juul	Missing industry
Carvana	Missing industry

Rather than assigning arbitrary categories, existing records for the same companies were examined.

```
SELECT
    company,industry
FROM raw_layoffs
WHERE
company IN ('Airbnb','Bally\'s Interactive','Juul','Carvana');
```

The query revealed the following information:

Company	Industry found in other records
Airbnb	Travel
Bally's Interactive	No usable industry value found in the returned records
Juul	Consumer
Carvana	Transportation

Based on these findings, the missing industry values for Airbnb, Juul, and Carvana were filled using existing information.

The following SQL statements were used:

```
UPDATE raw_layoffs
SET industry = 'Travel'
WHERE company = 'Airbnb'
  AND (industry IS NULL OR industry = '');

UPDATE raw_layoffs
SET industry = 'Consumer'
WHERE company = 'Juul'
  AND (industry IS NULL OR industry = '');

UPDATE raw_layoffs
SET industry = 'Transportation'
WHERE company = 'Carvana'
  AND (industry IS NULL OR industry = '');
```

For Bally's Interactive, Gambling was selected as a proposed industry classification based on the company's business activities.

```
UPDATE raw_layoffs
SET industry = 'Gambling'
WHERE company = 'Bally\'s Interactive'
  AND (industry IS NULL OR industry = '');
```

Result: Missing industry values were investigated using existing records and company information. The approach reduced missing information while avoiding changes to industry values that were already populated.

Step 9: Identify Missing Layoff Information

The next step focused on missing values in the two columns most directly related to layoff size:

- total_laid_off

- percentage_laid_off

The following query was executed:

```
SELECT *
FROM raw_layoffs
WHERE total_laid_off IS NULL
  AND percentage_laid_off IS NULL;
```

Finding: A total of 361 records had NULL values in both columns.

Metric	Count
Records in raw_layoffs	2,356
Records with both fields NULL	361

These records lacked both the absolute number of employees laid off and the percentage of the workforce affected.

However, the records could still contain useful information about companies, locations, industries, dates, and funding.

The records were therefore not deleted from raw_layoffs. Instead, a separate table was created containing only records with complete information in both layoff-related columns.

6. Creation of the Final Clean Dataset

A new table named clean_layoffs was created to store records where both total_laid_off and percentage_laid_off contained values.

The following query was used:

```
CREATE TABLE clean_layoffs AS
SELECT *
FROM raw_layoffs
WHERE total_laid_off IS NOT NULL
  AND percentage_laid_off IS NOT NULL;
```

This query created a new table and imported only the records satisfying both conditions.

The filtering conditions were important because a record could have one populated field while the other was NULL.

For example:

total_laid_off	percentage_laid_off	Included in clean_layoffs?
500	0.10	Yes
500	NULL	No
NULL	0.10	No
NULL	NULL	No

After executing the query, the resulting table contained **1,194 rows and nine columns**.

7. Final Dataset Summary

The following table summarizes the transformation from the original imported dataset to the final clean dataset.

Stage	Records	Description
Original imported dataset	2,361	Initial CSV import
After duplicate removal	2,356	Five excess duplicate records removed
Records with both layoff fields missing	361	Identified during missing-value analysis
Records with at least one layoff field missing	1,162	Excluded from the final table
Final clean_layoffs dataset	1,194	Both layoff-related fields populated

The final dataset contains approximately **50.68%** of the records in the deduplicated working dataset.

Calculation:

$$\text{Retention Rate} = \frac{1{,}194}{2{,}356} \times 100$$

$$\boxed{\text{Retention Rate } = 50.68\%}$$

The remaining 1,162 records were excluded from clean_layoffs because at least one of the two required layoff-related fields was missing.

These records remain in raw_layoffs, preserving the broader dataset for potential future investigation.

8. Key Findings and Observations

The data cleaning process revealed several important data quality issues.

1. Duplicate records

Five excess duplicate records were identified and removed. This reduced the working dataset from 2,361 to 2,356 records.

2. Inconsistent text values

Leading and trailing whitespace in company names, inconsistent cryptocurrency industry labels, and trailing periods in country names were identified and standardized.

3. Missing industry information

Missing industry values were investigated using other records belonging to the same companies. Existing information was used to fill missing values where suitable evidence was available.

4. Incorrect date representation

The date column was converted from text to the DATE data type, preparing it for reliable chronological analysis.

5. Missing layoff-related values

A significant number of records lacked either the number of employees laid off or the percentage of the workforce affected. Consequently, only 1,194 records satisfied the requirement that both values be present.

6. Separation of raw and clean data

The original imported table was preserved, and a working table was used for cleaning. A separate clean_layoffs table was then created for records meeting the selected completeness criteria.

This approach supports data traceability and allows the original information to be consulted if necessary.

9. Conclusion

This project demonstrates how MySQL can be used to identify and address common data quality problems in a real-world layoffs dataset.

The cleaning process involved duplicate detection and removal, text standardization, correction of inconsistent industry and country labels, date conversion, investigation of missing industry information, and filtering based on the completeness of layoff-related fields.

The original dataset contained 2,361 records. After removing five excess duplicate records, 2,356 records remained in the working dataset.

A separate clean_layoffs table was subsequently created with 1,194 records and nine columns. Each record in this table has a non-NULL value for both total_laid_off and percentage_laid_off.

The resulting dataset provides a more consistent foundation for subsequent exploratory data analysis and business intelligence tasks.

However, the final table represents a filtered subset rather than the entire layoffs dataset. Because records with either missing layoff-related value were excluded, future analysis using clean_layoffs should acknowledge this selection criterion.

Overall, the project establishes a structured data cleaning workflow that improves consistency and prepares the dataset for further analysis.

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10. Scope for Future Work

The project has currently reached the creation of the clean_layoffs table. Potential next steps include:

1. Validate the final table for remaining missing values and duplicate records.
2. Examine outliers and unusual values in the numerical columns.
3. Perform exploratory data analysis on layoffs by company, industry, country, and date.
4. Identify industries and companies with the largest reported layoffs.
5. Analyze layoff trends over time.
6. Develop visualizations and dashboards using Power BI.
7. Document any analytical limitations caused by excluding records with missing layoff-related information.

These activities would build upon the cleaned dataset without changing the scope of the data cleaning work completed so far.

Report status: Completed through the creation of the clean_layoffs table.